Lecture Plan

PHYSICS 2 [1203]: Credit: 3 [24 lectures (36 hours)]
Heat and Thermodynamics & Oscillations, Waves and Optics
Midterm Examination, Spring 2019-20

Text book: Fundamentals of Physics (10th edition, Extended): David Halliday, Robert Resnick and Jearl Walker (John Wiley & Sons)

Midterm Examination [40%]: Heat and Thermodynamics [12 lectures (18 hours)]

1st Lecture:
Description of the course

Objective:
Chapter 18: Temperature, heat and the first law of thermodynamics
18-1: Temperature (temperature, the zeroth law of thermodynamics), 18-4: Absorption of heat (temperature and heat), The absorption of heat by solids and liquids (heat capacity, specific heat, molar specific heat, heat of transformation)

Outcome:
1. Explain the zeroth law of thermodynamics.
2. Explain the heat capacity, specific heat capacity, molar specific heat, heat of transformation, heat of vaporization and heat of fusion.

2nd Lecture:

Application:
Module 18-4 Absorption of Heat
23. A small electric immersion heater is used to heat 100 g of water for a cup of instant coffee. The heater is labeled “200 watts” (it converts electrical energy to thermal energy at this rate). Calculate the time required to bring all this water from 23.0 ^{0}C to 100 ^{0}C, ignoring any heat losses.

24. A certain substance has a mass per mole of 50.0 g/mol. When 314 J is added as heat to a 30.0 g sample, the sample’s temperature rises from 25.0 ^{0}C to 45.0 ^{0}C. What are the (a) specific heat and (b) molar specific heat of this substance? (c) How many moles are in the sample?

27. Calculate the minimum amount of energy, in joules, required to completely melt 130 g of silver initially at 15.0 ^{0}C.

28. How much water remains unfrozen after 50.2 kJ is transferred as heat from 260 g of liquid water initially at its freezing point?

3rd Lecture:
Objective:

18-5: The first law of thermodynamics (a closer look at heat and work [one path, another path, reversed steps, negative work and cycle], the first law of thermodynamics), Some special cases of the first law of thermodynamics (adiabatic processes, constant-volume processes, cyclical processes and free expansions)

Outcome:

Chapter 18: Probable questions

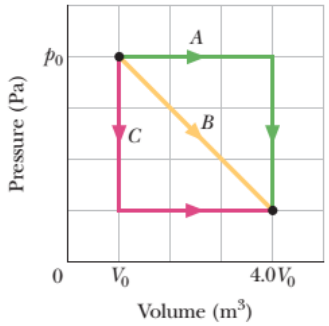
3. A gas may exchange energy with its surroundings at constant pressure through work. Find an expression for work done by any gas as it expands or contracts from an initial volume to a final volume.
4. Explain the first law of thermodynamics.
5. Apply the first law of thermodynamics for adiabatic processes, constant-volume processes, cyclical processes and free expansions.

4th Lecture:

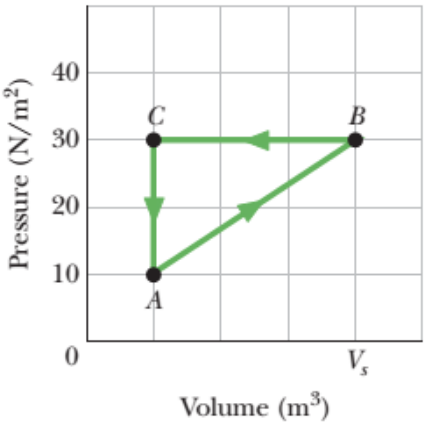
Application:

Module 18-5 The First Law of Thermodynamics

43. In Fig., a gas sample expands from V_0 to $4.0V_0$ while its pressure decreases from p_0 to $p_0/4.0$. If V_0 = 1.0 m^3 and p_0 = 40 Pa, how much work is done by the gas if its pressure changes with volume via (a) path A, (b) path B, and (c) path C?

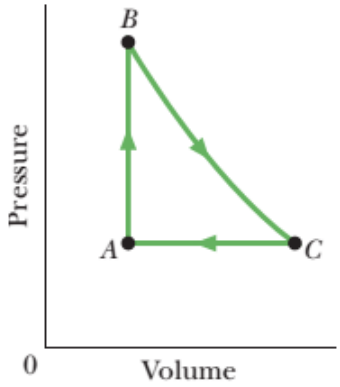


45. A gas within a closed chamber undergoes the cycle shown in the *p-V* diagram of Fig. The horizontal scale is set by V_s = 4.0 m^3. Calculate the net energy added to the system as heat during one complete cycle.



46. Suppose 200 J of work is done on a system and 70.0 cal is extracted from the system as heat. In the sense of the first law of thermodynamics, what are the values (including algebraic signs) of (a) W, (b) Q, and (c) ΔE_{int}?

48. As a gas is held within a closed chamber, it passes through the cycle shown in Fig. Determine the energy transferred by the system as heat during constant-pressure process CA if the energy added as heat Q_{AB} during constant-volume process AB is 20.0 J, no energy is transferred as heat during adiabatic process BC, and the net work done during the cycle is 15.0 J.



5th Lecture:
1st quiz

Objective:
Chapter 19: The kinetic theory of gases [4 lectures (6 hours), 2nd quiz]

19-1: Avogadro's number, 19-2: Ideal gases (ideal gases, work done by an ideal gas at constant temperature, work done at constant volume and at constant pressure)

Outcome:
Chapter 19:
1. Derive the ideal gas law, written in terms of the number of moles *n*.
2. Derive the work done by an ideal gas during an isothermal process, a constant-volume process and a constant-pressure process.

Application:
Module 19-2 Ideal Gases

4. A quantity of ideal gas at 10.0 ^{0}C and 100 kPa occupies a volume of 2.50 m^3. (a) How many moles of the gas are present? (b) If the pressure is now raised to 300 kPa and the temperature is raised to 30.0 ^{0}C, how much volume does the gas occupy? Assume no leaks.

7. Suppose 1.80 mol of an ideal gas is taken from a volume of 3.00 m^3 to a volume of 1.50 m^3 via an isothermal compression at 30 ^{0}C. (a) How much energy is transferred as heat during the compression, and (b) is the transfer *to* or *from* the gas?

6th Lecture:

Objective:
19-3: Pressure, Temperature, and RMS speed, 19-4: Translational kinetic energy

Outcome:
Chapter 19: Probable questions
3. For an ideal gas, derive the expression for rms speed as a function of temperature.
4. Derive the relationship between the average kinetic energy and the temperature of the gas.

Application:
Module 19-3 Pressure, Temperature, and RMS Speed
18. The temperature and pressure in the Sun's atmosphere are 2.00x10^6 K and 0.0300 Pa. Calculate the rms speed of free electrons (mass 9.11x10^{-31} kg) there, assuming they are an ideal gas.

23. A beam of hydrogen molecules (H_2) is directed toward a wall, at an angle of 55^0 with the normal to the wall. Each molecule in the beam has a speed of 1.0 km/s and a mass of 3.3×10^{-24} g. The beam strikes the wall over an area of 2.0 cm^2, at the rate of 10^{23} molecules per second. What is the beam's pressure on the wall?

7th Lecture:

Application:
Module 19-4 Translational Kinetic Energy

25. Determine the average value of the translational kinetic energy of the molecules of an ideal gas at temperatures (a) 0.00 ^{0}C and (b) 100 ^{0}C. What is the translational kinetic energy per mole of an ideal gas at (c) 0.00 ^{0}C and (d) 100 ^{0}C?

26. What is the average translational kinetic energy of nitrogen molecules at 1600 K?

Objective:
19-7: The molar specific heats of an ideal gas (internal energy, molar specific heat at constant volume, molar specific heat at constant pressure), 19-8: Degrees of freedom and molar specific heats

Outcome:
Chapter 19: Probable questions
5. Drive an expression for the internal energy of an ideal gas is a function of the gas temperature only.
6. If C_v and C_p are molar specific heats at constant volume and constant pressure respectively of an ideal gas, then show that $C_p - C_v = R$, where R is the gas constant. Why is C_p greater than C_v ?
7. Calculate the molar specific heat for monatomic, diatomic and polyatomic ideal gases in a constant-volume process and a constant-pressure process.

Application:
Module 19-7 The Molar Specific Heats of an Ideal Gas

47. The temperature of 2.00 mol of an ideal monatomic gas is raised 15.0 K at constant volume. What are (a) the work W done by the gas, (b) the energy transferred as heat Q, (c) the change ΔE_{int} in the internal energy of the gas, and (d) the change ΔK in the average kinetic energy per atom?

48. When 20.9 J was added as heat to a particular ideal gas, the volume of the gas changed from 50.0 cm^3 to 100 cm^3 while the pressure remained at 1.00 atm. (a) By how much did the internal energy of the gas change? If the quantity of gas present was 2.00×10^{-3} mol, find (b) C_p and (c) C_V.

8th Lecture:
2nd Quiz

Objective:
19-9: The adiabatic expansion of an ideal gas (the adiabatic expansion of an ideal gas and free expansions)

Outcome:
Chapter 19: Probable questions
8. Prove that the relation between the pressure and the volume during an adiabatic process of an ideal gas is pV^{γ} = a constant, where $\gamma = C_p/C_v$.

Application:
Module 19-9 The Adiabatic Expansion of an Ideal Gas

54. We know that for an adiabatic process pV^{γ} = a constant. Evaluate "a constant" for an adiabatic process involving exactly 2.0 mol of an ideal gas passing through the state having exactly p = 1.0 atm and T = 300 K. Assume a diatomic gas whose molecules rotate but do not oscillate.

55. A certain gas occupies a volume of 4.3 L at a pressure of 1.2 atm and a temperature of 310 K. It is compressed adiabatically to a volume of 0.76 L. Determine (a) the final pressure and (b) the final temperature, assuming the gas to be an ideal gas for which γ = 1.4.

9th Lecture:

Objective:

Chapter 20: Entropy and the second law of thermodynamics [4 lectures (6 hours)]
20-1: Entropy (irreversible processes and entropy, change in entropy, entropy as a state function, the second law of thermodynamics)

Outcome:
Chapter 20: Probable questions
1. Explain the reversible and irreversible processes. Explain entropy.
2. Explain the second law of thermodynamics.
3. When an ideal gas changes reversibly from an initial state with temperature T_i and volume V_i to a final state with temperature T_f and volume V_f, find the change in the entropy of the gas.

10th Lecture:
Application:
Chapter 20: Probable tutorial problems
Module 20-1 Entropy

2. An ideal gas undergoes a reversible isothermal expansion at 77.0 ^{0}C, increasing its volume from 1.30 L to 3.40 L. The entropy change of the gas is 22.0 J/K. How many moles of gas are present?

3. A 2.50 mol sample of an ideal gas expands reversibly and isothermally at 360 K until its volume is doubled. What is the increase in entropy of the gas?

4. How much energy must be transferred as heat for a reversible isothermal expansion of an ideal gas at 132 ^{0}C if the entropy of the gas increases by 46.0 J/K?

11th Lecture:

Objective:
20-2: Entropy in the real world: Engines (a Carnot engine, the work, entropy changes, efficiency of a Carnot engine)

Outcome:
4. Sketch a p-V diagram for the cycle of a Carnot engine, indicating the direction of cycling, the nature of the processes involved, the work done during each process (including algebraic sign), the net work done in the cycle, and the heat transferred during each process (including algebraic sign).
5. Sketch a Carnot cycle on a temperature–entropy diagram, indicating the heat transfers.
6. Determine the net entropy change around a Carnot cycle.
7. Derive the efficiency of a Carnot engine in terms of the heat transfers and also in terms of the temperatures of the reservoirs.
8. Explain that there are no perfect engines in which the energy transferred as heat, Q from a high temperature reservoir goes entirely into the work, W done by the engine.

12th Lecture:
3rd Quiz

Application:
Module 20-2 Entropy in the Real World: Engines
23. A Carnot engine whose low-temperature reservoir is at 17 ^{0}C has an efficiency of 40%. By how much should the temperature of the high-temperature reservoir be increased to increase the efficiency to 50%?

24. A Carnot engine absorbs 52 kJ as heat and exhausts 36 kJ as heat in each cycle. Calculate (a) the engine's efficiency and (b) the work done per cycle in kilojoules.

25. A Carnot engine has an efficiency of 22.0%. It operates between constant-temperature reservoirs differing in temperature by 75.0 C^0 .What is the temperature of the (a) lower-temperature and (b) higher-temperature reservoir?

27. A Carnot engine operates between 235 ^{0}C and 115 ^{0}C, absorbing 6.30x10^4 J per cycle at the higher temperature. (a) What is the efficiency of the engine? (b) How much work per cycle is this engine capable of performing?